



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
07-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [SCORE-III (ALL STAR BATCHES)]

PAPER-1

Time : 3 Hours

Maximum Marks : 210

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 15)

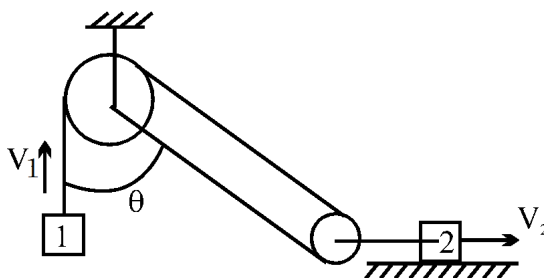
- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. In the arrangement shown in figure, the ratio of velocity V_1 & V_2 of block (1) & (2) is :-

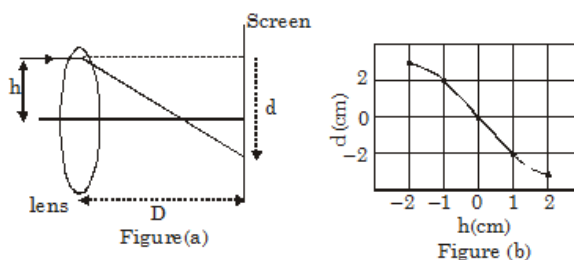


- (A) $\frac{V_1}{V_2} = 2 \cot \theta$ (B) $\frac{V_1}{V_2} = \frac{\sin \theta}{2}$ (C) $\frac{V_1}{V_2} = 2 \sin \theta$ (D) None
2. A long cylinder of radius 'a' is made of a material whose thermal conductivity depends on distance 'r' from axis of the conductor as $K = cr^2$, where 'c' is a constant. Considering heat flow along the length, find thermal resistance per unit length of such a conductor.

- (A) $\frac{1}{\pi ca^4}$ (B) $\frac{2}{\pi ca^4}$ (C) $\frac{2}{\pi ca^3}$ (D) $\frac{1}{2\pi ca^4}$

Space for Rough Work

3. The vertical displacement of light rays parallel to the axis of a lens is measured as a function of the vertical displacement h of the incident ray from the principal axis as shown in figure (a). The data is plotted in figure (b). The distance D from the lens to the screen is 1.0 m. Find the focal length of the lens (in cm) (for paraxial rays)



- (A) 30 (B) 50 (C) $\frac{200}{3}$ (D) $\frac{100}{3}$
4. A wall of width ℓ and cross sectional area A is having variable coefficient of thermal conductivity given by $k = k_0 + \alpha x$ (where k_0 and α are positive constant) and x is measured from outer surface of wall. If the temperature of surroundings is T_0 °C and the temperature of the room is maintained at T_r °C ($T_0 > T_r$), the temperature of the wall at $x = \frac{\ell}{2}$ is (given $\frac{\alpha \ell}{K_0} = 1$ S.I. unit, $T_0 = 45^\circ\text{C}$, $T_r = 20^\circ\text{C}$, $\ln 3 = 1.1$ and $\ln 2 = 0.69$)
- (A) 20°C (B) 25°C (C) 28°C (D) 30.5°C
5. Polychromatic light described at a place by the equation
 $E = 100 \sin(0.5\pi \times 10^{15}t) + \cos(\pi \times 10^{15}t) + \sin(2\pi \times 10^{15}t)$
 where E is in V/m and t in sec, falls on a metal surface having work function 2.0 eV. The maximum kinetic energy of the photoelectron is [Take $h = \text{Planck's constant} = 6.4 \times 10^{-34}$ J-s]
- (A) zero (B) 1 eV (C) 2 eV (D) 3 eV

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

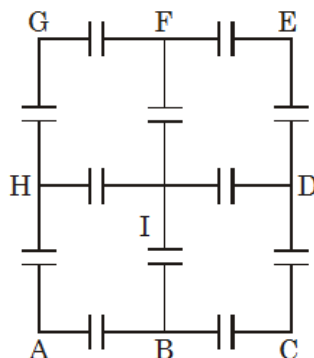
6. Energy liberated in the de-excitation of hydrogen atom from 3rd level to 1st level falls on a photo-cathode. Later when the same photo-cathode is exposed to a spectrum of some unknown hydrogen like gas, excited to 2nd energy level, it is found that the de-Broglie wavelength of the fastest photoelectrons, now ejected has decreased by a factor of 3. For this new gas, difference of energies of 2nd Lyman line and 1st Balmer line is found to be 3 times the ionization energy of the hydrogen atom. Select the correct statement(s):
- (A) The gas is lithium
- (B) The gas is helium
- (C) The work function of photo-cathode is 8.5 eV
- (D) The work function of photo-cathode is 5.5 eV

Space for Rough Work

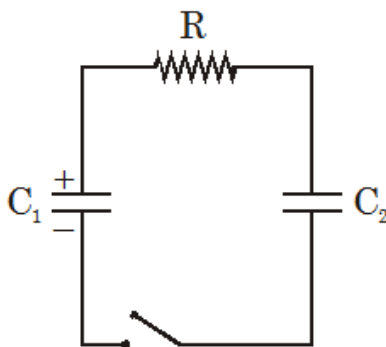
7. An air bubble in a water tank rises from the bottom to the top. Which of the following statements are true ?
- (A) Bubble rises upwards because pressure at the bottom is less than that at the top.
- (B) Bubble rises upwards because pressure at the bottom is greater than that at the top.
- (C) As the bubble rises, its size increases.
- (D) As the bubble rises, its size decreases.
8. A gaseous mixture enclosed in a vessel of volume consists of one mole of a gas A with $\gamma (= C_p/C_v) = 5/3$ and another gas B with $\gamma = 7/5$ at a certain temperature T. The molar masses of gases A and B are 4gm and 32gm respectively. The gases A and B do not react with each other and are assumed to be ideal. The gaseous mixture follows the equation $PV^{19/13} = \text{constant}$, in adiabatic process.
- (A) The number of moles of the gas B in the gaseous mixture is 2 mole
- (B) The speed of sound in this mixture at 300 K is 401 m/s (approx.)
- (C) If T is raised by 1 K from 300 K, the percentage change in the speed of sound is 1/6
- (D) If T is raised by 1 K from 300 K, the percentage change in the speed of sound is 1/5

Space for Rough Work

9. Consider capacitance circuit consisting 12 identical capacitors each of capacitance $1\mu\text{F}$ as shown in the figure. The equivalent capacitance between the terminals



- (A) A and E is $\frac{2}{3}\mu\text{F}$ (B) A and I is $\frac{8}{7}\mu\text{F}$
 (C) H and I is $\frac{12}{7}\mu\text{F}$ (D) H and B is $\frac{5}{7}\mu\text{F}$
10. In the circuit shown in the figure $C_1 = C_2 = C$ and capacitor C_1 is having initial charge Q . The switch is closed at $t = 0$. Which of the following option(s) is(are) correct



- (A) The current at time t through the circuit will be $\left(\frac{Q}{RC}\right)e^{-2t/RC}$
 (B) The charge on capacitor C_1 at time t will be $\frac{Q}{2}\left(1 + e^{-t/RC}\right)$
 (C) The charge on the capacitor C_2 at time t will be $\frac{Q}{2}\left(1 - e^{-2t/RC}\right)$
 (D) Total heat loss in the process of charge transfer will be $\frac{Q^2}{4C}$

Space for Rough Work

11. A spherical ball of radius R carries a charge Q which is distributed in its volume in a spherically symmetric manner. The surrounding space is filled with a charge of volume density $\rho = \beta/r$ where β is a constant and r is the distance from the centre of the ball. The relative permittivity of the ball and the surrounding space are assumed to be equal to unity. Then the charge Q of the ball for which the magnitude of the electric field strength (outside the ball) is independent of r and the magnitude of electric field strength (E) outside ball is:

(A) $Q = 2\pi\beta R^2$

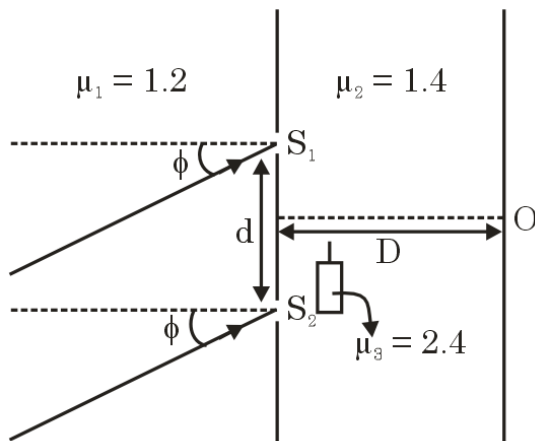
(B) $Q = 4\pi\beta R^2$

(C) $E = \frac{4\beta}{\epsilon_0}$

(D) $E = \frac{\beta}{2\epsilon_0}$

Space for Rough Work

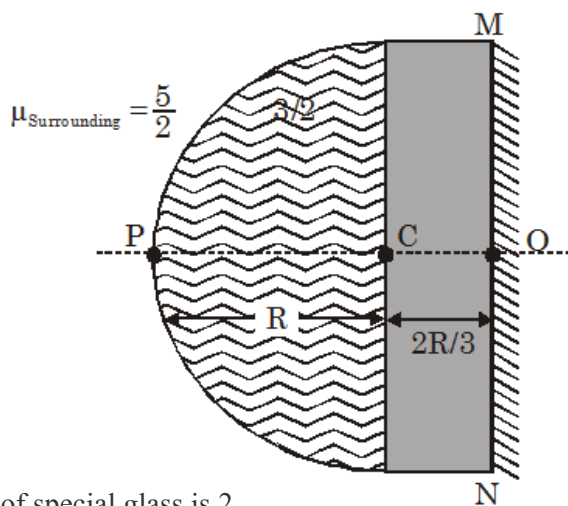
12. In a YDSE shown in the figure, a parallel beam of light of wavelength $5000\text{\AA}$ (in air) and intensity I , travelling in medium $\mu_1 = 1.2$ is incident on slits S_1 and S_2 at an angle $\phi = 30^\circ$. On the other side of slits medium $\mu_2 = 1.4$ is present. A thin transparent slab of thickness $t = 0.6\text{ mm}$ and refractive index $n_3 = 2.4$ is placed in front of S_2 . Slits are $d = 1\text{ mm}$ apart and distance between screen and plane of slits is $D = 1\text{ m}$ and O is the centre of screen.



- (A) The position of central maxima is at 'O'
- (B) The angular position relative to central line of central maxima if the slab of $\mu_3 = 2.4$ is removed is, $\sin^{-1}\left(\frac{3}{7}\right)$
- (C) Fringe width on the screen is $\frac{5}{14}\text{ mm}$ (with slab of $\mu_3 = 2.4$ present)
- (D) Intensity at 'O' is $4I$ (with slab of $\mu_3 = 2.4$ present) (I = intensity of light on the screen from both slits)

Space for Rough Work

13. A solid hemisphere of refractive index $\frac{3}{2}$ and radius $R = 20$ cm is placed as shown in the figure. Another special cylindrical glass of refractive index ' μ ', radius R and thickness $\frac{2R}{3}$ is also placed adjacent to flat surface of solid hemisphere. There is a very small luminous spot at 'P' just inside the solid hemisphere and MN surface of special glass surface is silvered. The position of the final image of point 'P' for an observer, left of hemisphere is at point (O). (Consider only paraxial rays).



- (A) The refractive index (μ) of special glass is 2
- (B) The refractive index (μ) of special glass is $\frac{3}{2}$
- (C) An observer is located inside the special cylindrical glass medium then the distance between the images as observed by that observer, is 80 cm.
- (D) An observer is located inside the special cylindrical glass medium then the distance between the images as observed by that observer, is $80/3$ cm.

Space for Rough Work

14. The potential energy of a particle of mass 0.1 kg, moving along the x-axis, is given by $U = 5x(x - 4)$ J, where x is in metres. It can be concluded that :
- (A) the particle is acted upon by a constant force.
(B) the speed of the particle is maximum at $x = 2$ m.
(C) the particle executes simple harmonic motion
(D) the period of oscillation of the particle is $\pi/5$ s.
15. A concave spherical mirror has a radius of curvature of 50 cm. Find two positions of an object for which the image is four times as large as the object.
- (A) $\frac{75}{4}$ cm from mirror (B) 125 cm from mirror
(C) $\frac{125}{4}$ cm from mirror (D) 32 cm from mirror
-

Space for Rough Work

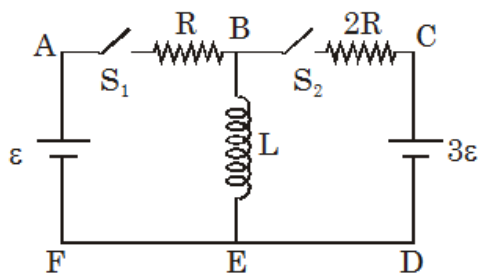
SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

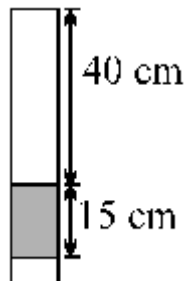
Zero Marks : 0 In all other cases.

1. A radioactive source is a mixture of two radioactive materials X_1 and X_2 with mean life-time of 6 days and 9 days respectively. Currently 20% of radiations came from the radioactive decay of X_1 and remaining by X_2 . The time in past at which 80% of radiations came from the decay of X_1 is given by $t = 9n \ln 16$ days. What is the value of n ?
2. In the figure shown switch S_1 remains connected for a long time and the switch S_2 was open. Now the switch S_2 is also closed. Assuming $\varepsilon = 10$ V and $L = 1$ H. The magnitude of rate of change of current in inductor (A/s) just after the switch S_2 is closed, is $\frac{30}{N}$. Find N



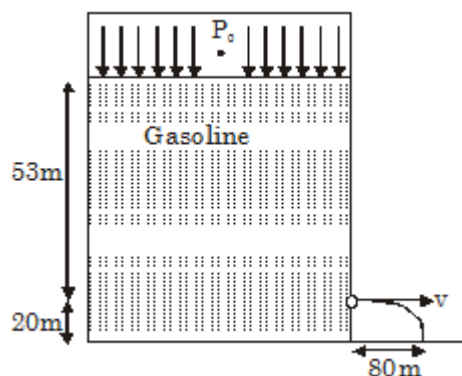
Space for Rough Work

3. A certain amount of air at 300 K is trapped in a glass tube between its closed end and a 15 cm long mercury column as shown in figure. Other end of tube is open to atmosphere. Tube is now inverted isothermally due to which volume of air decreases. If the air column is heated to the temperature (T in K) so that it regains its original volume, find value of $\frac{T}{10}$. (Atmospheric pressure = 75 cm of Hg)



Space for Rough Work

4. A tank containing gasoline is sealed and the gasoline is under pressure P_0 as shown in the figure. The stored gasoline has a density of 660 kgm^{-3} . A sniper fires a rifle bullet into the gasoline tank, making a small hole 53 m below the surface of gasoline. The total height of gasoline is 73 m from the base. The jet of gasoline shooting out of the hole strikes the ground at a distance of 80 m from the tank initially. The outside atmospheric pressure is 10^5 Nm^{-2} . If the pressure above the gasoline surface is $P_0 = (a \times 10^b) \text{ Nm}^{-2}$, then $(a + b)$ is equal to [Where a and b are integers between 0 and 9. Use standard notation and rounding off] [$g = 10 \text{ m/s}^2$]



5. A point A is located at a distance $r = 1.5 \text{ m}$ from a point source of sound of frequency $n = 600 \text{ Hz}$. The power of the source $P = 0.80 \text{ W}$. Neglecting the damping of the wave and assuming the velocity of the sound in air to be equal to $V = 340 \text{ ms}^{-1}$. Find the pressure oscillation amplitude $(\Delta p)_m$ (in SI units) at the point A (Use $d_{\text{air}} = \frac{225\pi}{544} \text{ kgm}^{-3}$)

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I(i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

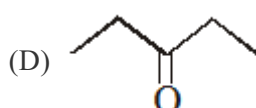
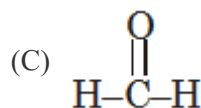
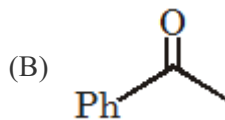
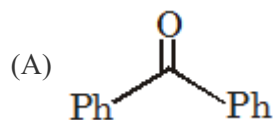
- The correct order of elements having second ionization energies of 1757, 5247, 2423 and 7293 kJ mol⁻¹, respectively, is
(A) Be, Li, He, B (B) Be, Li, B, He (C) Be, He, B, Li (D) B, He, Li, Be
- Which form of sulphur is formed on evaporation of the solution of roll sulphur in CS₂.
(A) Amorphous sulphur (B) Rhombic sulphur
(C) Monoclinic sulphur (D) Cyclo-S₆ sulphur
- Select the correct statement for fixed moles of ideal gas undergoing from same initial state to same final volume
(A) $(\Delta S_{\text{surr.}})_{\text{rev adiabatic expansion}} > (\Delta S_{\text{surr.}})_{\text{rev isothermal expansion}}$
(B) $(\Delta S_{\text{surr.}})_{\text{rev adiabatic expansion}} > (\Delta S_{\text{surr.}})_{\text{irr. adiabatic expansion}}$
(C) $(\Delta S_{\text{surr.}})_{\text{rev isothermal expansion}} = (\Delta S_{\text{surr.}})_{\text{irr. isothermal expansion}}$
(D) $(\Delta S_{\text{surr.}})_{\text{irr. adiabatic compression}} > (\Delta S_{\text{surr.}})_{\text{irr. isothermal compression}}$

Space for Rough Work

4. The conductance of 0.0015M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinised Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1cm^2 . The conductance was found to be $5 \times 10^{-7}\text{S}$. The pH of the solution is 4. The value of limiting molar conductivity of this weak monobasic acid in aq. solution is :-

- (A) $600\text{ S cm}^2\text{ mol}^{-1}$ (B) $400\text{ S cm}^2\text{ mol}^{-1}$
(C) $300\text{ S cm}^2\text{ mol}^{-1}$ (D) $500\text{ S cm}^2\text{ mol}^{-1}$

5. Compound which gives positive iodoform test is :



Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

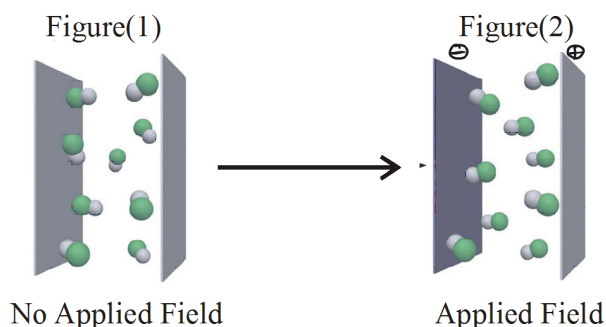
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. In an experiment vapours of compound (X) are placed between pair of metal plates in the absence of applied field in figure(I) and in the presence of applied electrical field when an electrical potential is applied across the plates in figure-2. Depends on compound(X) identify correct statement(s).



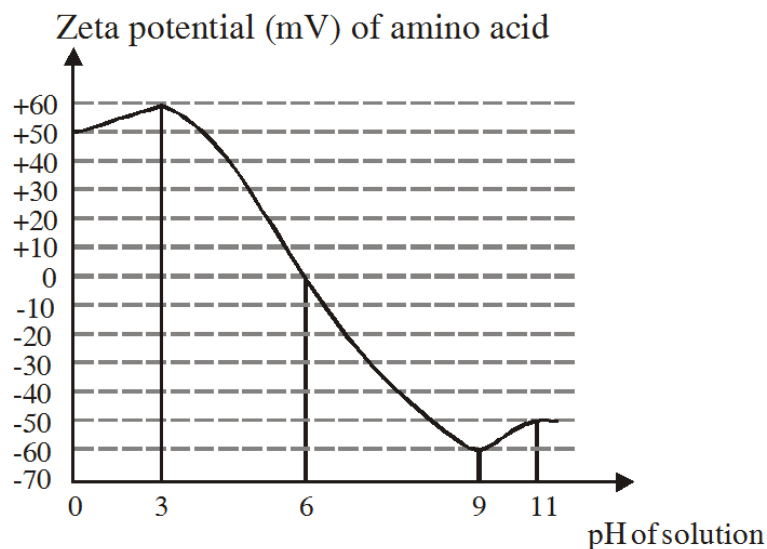
- (A) If X is NH_3 , N end of molecule is point towards +ve plate
- (B) If X is NF_3 , N end of molecule is point towards +ve end of plate
- (C) If X is BF_3 , No such alignment of molecules is observe
- (D) Extent of alignment is higher when (X) is NH_3 as compared to when (X) is NF_3

Space for Rough Work

7. Which of the following statement is/are **CORRECT** about metallurgical process?
- (A) In the Bayer process conc. alumina is extracted from aluminium ores in the mixture of TiO_2 , Al_2O_3 , $\text{Al}(\text{OH})_3$ & Fe_2O_3 .
- (B) Aluminium ion is reduced at a cathode from the mixture of molten cryolite and molten Al_2O_3 in Hall-Heroult process.
- (C) In the hydrometallurgical extraction of copper, leached Cu^{2+} ions from low grade ores is reduced by H_2 .
- (D) In the pyrometallurgical process, copper is reduced by self reduction from Cu_2S in silica lined convertor.
8. Select correct order of solubility :-
- (A) $\text{MnS} > \text{CdS}$ (B) $\text{K}_2\text{S} > \text{ZnS}$
- (C) $\text{HgCl}_2 > \text{Hg}_2\text{Cl}_2$ (D) $\text{PbCl}_2 > \text{SnCl}_2$
-

Space for Rough Work

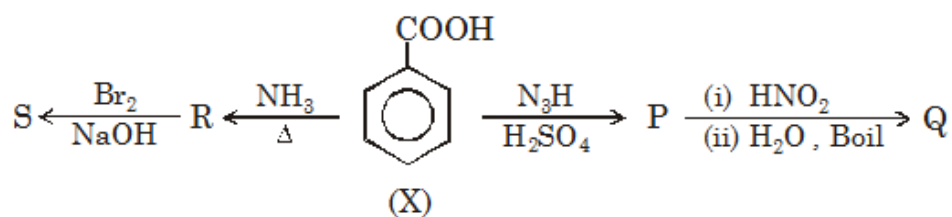
9. From above experimental graph, select the correct statement(s)



- (A) Amino acid is least stable at pH = 6
- (B) Amino acid has maximum stability at pH = 3 and 9
- (C) Amino acid is stable only when pH > 6.
- (D) Amino acid is stable only when pH < 6.
10. 0.768 gm of salt Na_2X is added to water to prepare 50 ml aq. solution. The solution is titrated with V ml 0.1 M H_2SO_4 aq. solution. If molar mass of weak acid ' H_2X ' is 52 gmol^{-1} & $[\text{K}_{\text{a}1}(\text{H}_2\text{X}) = 10^{-5}, \text{K}_{\text{a}2}(\text{H}_2\text{X}) = 10^{-8}]$. Identify correct option(s) for the solution formed. [Given : $\log 2 = 0.3, \log 3 = 0.48$]
- (A) If V = 20 then pH of final solution is 8
- (B) If V = 50 then pH of final solution is 5.48
- (C) If V = 80 then pH of final solution is 9.52
- (D) If V = 100 then pH of final solution is nearly 1.58

Space for Rough Work

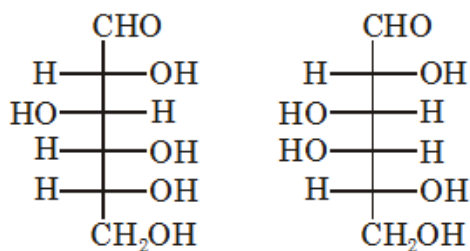
11. Which of the following statement(s) is/are true for a single electron atom/ion ?
- (A) Magnitude of orbital angular momentum of electron can be determined accurately but its direction cannot be determined accurately
- (B) Orbital angular momentum of electron has infinite possible directions in an atom/ion.
- (C) With respect to a fixed direction orbital angular momentum of electron in an orbital can have $(2l + 1)$ possible directions with each direction represented by each value of 'm'
- (D) Magnitude of spin angular momentum of electron is $\left(\sqrt{3} \frac{h}{2\pi}\right)$
12. For the following sequence, choose the correct option(s) ?



- (A) (Q) will give positive Br_2 , H_2O test
- (B) (R) on reaction with NaNO_2 , HCl produces (X)
- (C) Major product (P) & (S) are identical
- (D) Reaction sequence involves schmidt & Hoffmann bromamide degradation reaction

Space for Rough Work

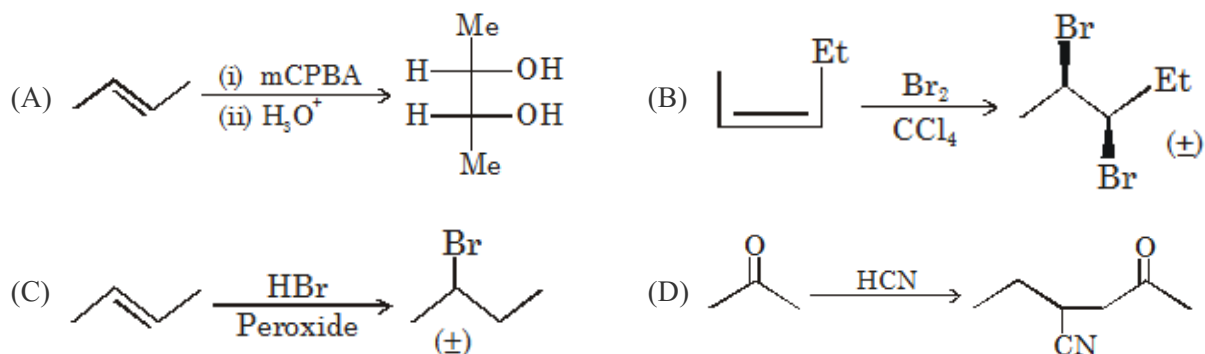
13. Following are the structure of D-Glucose and D-Galactose.



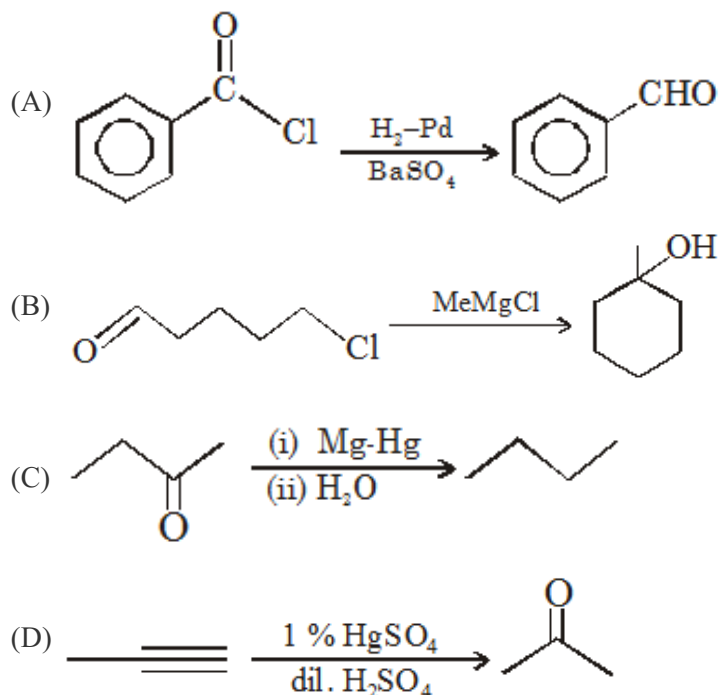
Which of the following statements are correct about these compounds ?

- (A) They are diastereomers (B) Both are component of lactose
(C) They are C-4 epimer (D) Both are optically active

14. Identify reactions correctly matched with their major product ?



15. Select the reaction(s) correctly match with their major product :



Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

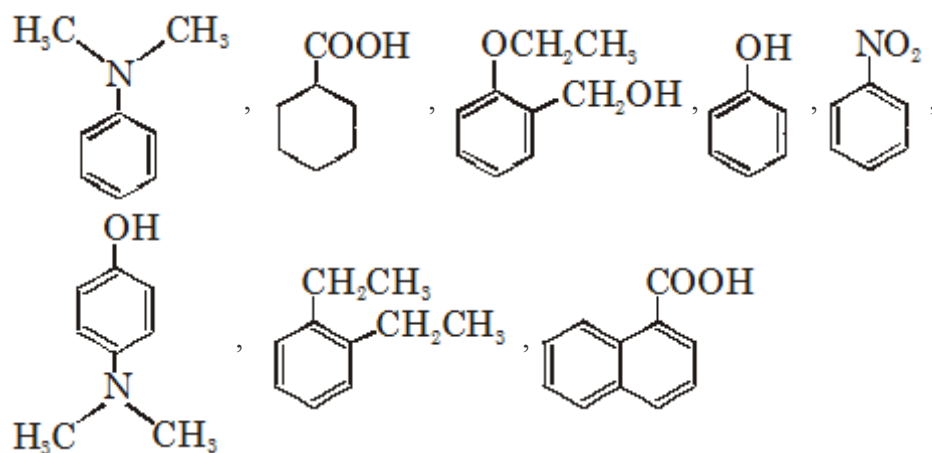
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Consider given reactions and find sum of molecular weight of Gas-1, Gas-3 & Gas-4.
(Atomic mass : H = 1, F = 19, N = 14, O = 16, C = 12, S = 32, Cl = 35.5, Br = 80, I = 127, P = 31)
- (i) $\text{HOCl} \xrightarrow[\text{(Catalytic decomposition)}]{\text{Co}^{+2}} \text{Gas-1} + \text{Gas-2}$
- (ii) Binary compound of metal $\xrightarrow{\Delta} \text{Gas-1} + \text{Solid-1}$
- (iii) $\text{Solid-1} \xrightarrow{\text{NaOH(aq.)}} \text{Solution-1} + \text{No gases product}$
 $\quad \quad \quad \downarrow \text{dil. HNO}_3 \quad \quad \quad \rightarrow \text{Solution-2} + \text{No gases product}$
- (iv) $\text{Gas-1} + \text{Gas-2} \xrightarrow[\text{with CuCl}_2]{\Delta} \text{Gas-3} + \text{other products}$
- (v) $\text{Solution-2} \xrightarrow{\text{Gas-2}} \text{white ppt}$
 $\quad \quad \quad \downarrow \text{H}_2\text{S} \quad \quad \quad \rightarrow \text{Black ppt}$
- (vi) Salt obtain by vapourization of solution-2 $\xrightarrow[\Delta]{\text{decomposition}} \text{Solid-1} + \text{Gas-1} + \text{Gas-4}$
(coloured)

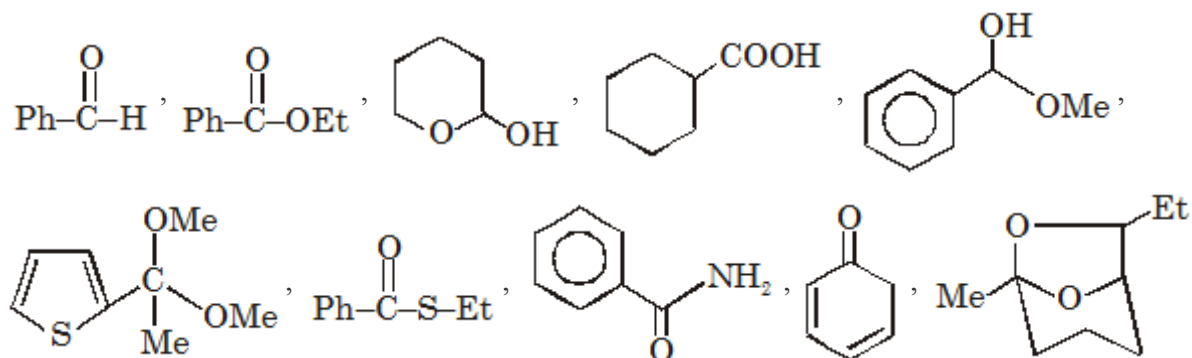
Space for Rough Work

2. Find sum of unpaired electrons in :-
Atomic Nitrogen ; $K_3[Fe(CN)_6]$; unstable S_2 molecule ; BaO_2 ; Nitric oxide ; Ozone ; Dioxygen ; $[Ti(H_2O)_6]Cl_3$; $[MnCl_6]^{3-}$; $[Mn(CN)_6]^{3-}$
3. Amongst the following, the total number of compounds soluble in aqueous $NaHCO_3$ is:



Space for Rough Work

4. The total number of compounds (shown below) that form phenyl hydrozone derivatives under acidic conditions is :



5. Two liquids A ($P_A^0 = 100\text{torr}$) & B ($P_B^0 = 225\text{torr}$) form ideal liquid solution, total pressure in torr over the liquid solution when $y_A - x_A$ is maximum. Where y_A and x_A have usual meaning.

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I(i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. The number of points of intersection of curve $\sin x = \cos y$ and $x^2 + y^2 = 1$, is-
 (A) 3 (B) 2
 (C) 0 (D) infinitely many
2. Let $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$ where $\vec{a} \wedge \vec{b} = 60^\circ$ & $\vec{c}$ bisects the angle between $\vec{a}$ & $\vec{b}$ and $|\vec{a} + \vec{b} + \vec{c}|^2 = (\ell + m\sqrt{3})$ where $\ell, m \in \mathbb{N}$ then $(\ell + m)$ is equal to
 (A) 5 (B) 15 (C) 25 (D) 35
3. Let $y = f(x)$ be a differentiable function and satisfy the differential equation $(f(x) + f'(x))f'(x) + (f'(x) + f''(x))f(x) = 1$, then general solution of the differential equation is-
 (A) $y^2 = k_1 e^{-2x} + x + k_2$ (B) $y^3 = k_1 e^{-2x} + x + k_2$
 (C) $y^2 = k_1 e^{2x} + x^2 + k_2$ (D) $y^3 = k_1 e^{-2x} + x^2 + k_2$

Space for Rough Work

4. Let $A(z_1)$, $B(z_2)$ and $C(z_3)$ be complex numbers, satisfying the equation $|z| = 1$ and also satisfying the relation $3z_1 = 2z_2 + 2z_3$. Then $|z_2 - z_3|$ is -
- (A) $\frac{\sqrt{7}}{2}$ (B) $\frac{\sqrt{7}}{4}$ (C) $\frac{\sqrt{7}}{6}$ (D) $\frac{\sqrt{7}}{8}$
5. Let S be the set of points $\{(x,y,z) \mid 0 \leq x, y, z \leq 3, x,y,z \in \mathbb{I}\}$. Three distinct points are randomly chosen from S . If the chosen points are (x_1, y_1, z_1) , (x_2, y_2, z_2) & (x_3, y_3, z_3) , then the probability that $x_1 < x_2 < x_3$, $y_1 \geq y_2 \geq y_3$, $z_1 \geq z_2 \geq z_3$ with $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right) \in S$ is-
- (A) $\frac{2}{651}$ (B) $\frac{11}{2976}$ (C) $\frac{13}{2976}$ (D) $\frac{17}{2976}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. Consider points $A(0,0,0)$, $B(1,1,1)$ and plane $P : x - y + z = 3$. Let $Q(a,b,c)$ be a moving point on plane P then identify the correct statement(s)
- (A) If $QA + QB$ is minimum then $a - c - 10b$ is equal to 1
- (B) If $QA + QB$ is minimum then $a - c - 10b$ is equal to 2
- (C) If $|QA - QB|$ is maximum then $a - 2b + c$ is equal to 0
- (D) If $|QA - QB|$ is minimum then Q is unique
7. If the ratio of 6th term from beginning and 6th term from end in the expansion of $(x^3 + \sqrt{2} \cdot x^{-2})^{15}$ is $\frac{4\sqrt{2}}{243}$, then the value of x is less than -
- (A) 2 (B) 1 (C) $(\sqrt{2})^{1/5}$ (D) $(\sqrt{3})^{1/5}$

Space for Rough Work

8. A hyperbola intersects an ellipse $x^2 + 9y^2 = 9$ orthogonally. The eccentricity of the hyperbola is reciprocal of that of ellipse. If the axes of the hyperbola are along coordinate axes, then
- (A) vertices of hyperbola are $\left(\pm \frac{8}{3}, 0\right)$
- (B) y coordinate of point of intersection of ellipse and hyperbola is either $\frac{1}{3}$ or $-\frac{1}{3}$
- (C) latus rectum of hyperbola is $\frac{2}{3}$
- (D) latus rectum of hyperbola is $\frac{4}{3}$
9. A bag contains 100 articles of which 90 are good and 10 are defective. The articles are tested one by one till all defective are obtained. Which of the following options are correct ?
- (A) Probability that first defective article is obtained in second testing is $\frac{1}{11}$
- (B) The probability that a particular defective article is obtained in 87th testing is $\frac{1}{100}$
- (C) Given that second article is defective the probability that first article was good is $\frac{10}{11}$
- (D) Given that second article is defective the probability that first article was defective is $\frac{1}{11}$
10. If the series of natural numbers is divided into groups (1), (2,3,4,5,6,7,8,9), (10,11,12,13,14,.....,36), and so on, then -
- (A) total number of natural numbers in first n groups is $\left(\frac{n(n+1)}{2}\right)^2$
- (B) first term in nth group is $\frac{1}{4}(n-1)^2n^2 + 1$
- (C) last term in nth group is $\frac{1}{4}(n-1)^2n^2 + n^3$
- (D) the sum of the numbers in nth group is $\frac{n^3}{4}(n^4 + n^2 + 2)$

Space for Rough Work

11. Let $f(x)$ be an even quadratic polynomial such that $-4 \leq f(1) \leq -1$ & $-1 \leq f(2) \leq 5$. Which of the following statements is true ?
- (A) Leading coefficient of $f(x)$ is $\frac{f(2) - f(1)}{3}$
- (B) $f(0) = \frac{f(2) - 4f(1)}{3}$
- (C) Maximum value of $f(3)$ is 20
- (D) Maximum value of $f(3)$ is 12
12. In $\triangle ABC$, if $\angle B = \sec^{-1}\left(\frac{5}{4}\right) + \operatorname{cosec}^{-1}\sqrt{5}$, $\angle C = \operatorname{cosec}^{-1}\left(\frac{25}{7}\right) + \cot^{-1}\left(\frac{9}{13}\right)$ and $c = 3$.
(All symbols used have their usual meaning in a triangle.) which of the following is/are correct
- (A) $\tan A, \tan B, \tan C$ are in A.P.
- (B) $\tan A, \tan B, \tan C$ are in G.P.
- (C) The distance between orthocentre and centroid of triangle with sides $a^2, b^{\frac{4}{3}}$ and c is equal to $\frac{5}{3}$
- (D) The distance between orthocentre and centroid of triangle with sides $a^2, b^{\frac{4}{3}}$ and c is equal to $\frac{10}{3}$

Space for Rough Work

13. If $P = \frac{1}{n^4} \prod_{r=1}^{2n} (n^2 + r^2)^{\frac{1}{n}}$, then $\lim_{n \rightarrow \infty} \ln P$ is equal to-

(A) $\int_0^2 \ln(1 + x^2) dx$

(B) $\int_0^1 \ln(x^2 - 4x + 5) dx$

(C) $\int_0^2 \ln(x^2 - 4x + 5) dx$

(D) $2 \int_0^1 \ln(1 + x^2) dx$

14. The complex numbers $z_1, z_2, \dots, z_n$, represents the vertices of a regular polygon of n sides in order, inscribed in a circle of unit radius and $z_3 + z_n = A z_1 + \overline{A} z_2$, which of the following statement/s is/are correct

(A) if $n = 4$ then $A = \sqrt{5}$

(B) if $n = 6$ then $A = \sqrt{4}$

(C) if $n = 8$ then $A = \sqrt{3}$

(D) if $n = 12$ then $A = \sqrt{2}$

Space for Rough Work

15. Let $f(x)$ and $g(x)$ are polynomials of degree 3, where $g(\alpha) = g'(\alpha) = 0$, $g''(\alpha) \neq 0$ and $\lim_{x \rightarrow \alpha} \frac{f(x)}{g(x)} = 0$. If $h(x) = f(x)g'(x) + f'(x).g(x)$ then which of the following is/are true -
- (A) Number of solution(s) which are common to all $f(x) = 0$, $g(x) = 0$, $h(x) = 0$ is 1.
- (B) Number of solution(s) which are common to all $f(x) = 0$, $g(x) = 0$, $h(x) = 0$ is 2.
- (C) Number of distinct roots of equation $h(x) = 0$ is 2.
- (D) $h'''(\alpha) = 0$

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $k = \int_{-2021}^{2021} \frac{dx}{1 + x^{2019} + \sqrt{1 + x^{4038}}}$ then the value of $\frac{k}{100}$ is
2. Let $y = f(x) = x^3 + x^2 + x + 1$, then the value of $\int_{-2}^1 f(x)dx + \int_1^4 f^{-1}(y) dy$ is-
3. If L.C.M. of (a, b) = $5^2 \times 3^4$, L.C.M. of (b, c) = $5^3 \times 3^4$ & L.C.M. of (c, a) = $5^3 \times 3^4$, then possible ordered triplets of (a,b,c) are k then $\frac{k}{5}$ is equal to

Space for Rough Work

4. If the probability that randomly selected point inside area bounded by parabola $y^2 = 8x$ and its latus rectum lies inside the ellipse $\frac{x^2}{2} + \frac{y^2}{16} = 1$ is $\frac{\sqrt{2}(a\pi + b)}{32}$, then value of $(a + b)$ is
5. In a tetrahedron OABC, G is the centre of the tetrahedron, P, Q and R are the mid points of OA, OB and OC respectively. If the shortest distance between the line GR and PQ is $\frac{1}{2k}$, then the value of k is (where O is origin, $\vec{OA} = \hat{i}$, $\vec{OB} = \hat{j}$ and $\vec{OC} = \hat{k}$)

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work